

# Alexandre St-Aubin

1225 W 14<sup>th</sup> Avenue, unit 305  
Vancouver, BC, Canada, V6H 1P7

alex.staubin2@gmail.com  
Personal website: <https://alestaubin.github.io>

## Education

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### The University of British Columbia

September 2025 – Present

*Master of Science in Computer Science*

*Vancouver, British Columbia, Canada*

- Supervised by Professor Evan Shelhamer, working on Deep Learning with a focus on Vision, Robustness, and Adaptation.

### McGill University

Fall 2022 – Winter 2025

*Joint Honours BSc, Mathematics and Computer Science, GPA: 3.88/4.0*

*Montréal, Québec, Canada*

- Graduated with Distinction and First Class Honours.
- Relevant Courses: Applied Machine Learning, Applied Robotics, Fundamentals of Computer Vision, Honours Convex Optimization, Topics in AI for Robotics, Reinforcement Learning.

## Research Experience

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**Research Interests:** Computer Vision, Deep learning, Robotics, Test-Time Adaptation, Robustness

### Research Intern

July 2024 – December 2024

*Mila - Quebec AI Institute, Complex Data Lab*

- Worked under the supervision of Professors Reihaneh Rabbany (McGill) and Jean-François Godbout (Université de Montréal).
- Conducted research on the societal and political impact of deepfakes, with a focus on the believability of AI-generated media.
- Evaluated state-of-the-art diffusion models to assess the quality and realism of generated images.

### Research Intern

May 2024 – September 2024

*McGill University, Centre for Intelligent Machines*

- Conducted research in long-horizon motion control of robotic arms, supervised by Professor Hsiu-Chin Lin at McGill University.
- Developed a novel method for single-shot learning of long-horizon manipulation tasks.
- St-Aubin, A., Abyaneh, A. and Lin, H.-C., Single-Shot Learning of Stable Dynamical Systems for Long-Horizon Manipulation Tasks.

### Directed Reading Program

January 2024 – May 2024

*McGill University, Faculty of Science, Dept. of Mathematics and Statistics*

- Engaged in a machine learning study and conducted a research project, guided by a graduate student.
- Report: *An Introduction to Supervised Machine Learning*.

## Awards & Honors

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### NSERC Canada Graduate Scholarship-Master's

2025 – 2026

*Held at the University of British Columbia.*

*Amount: 27,000 CAD*

### (Offer rejected) University of Waterloo financial support

2025 – 2026

*Master of Mathematics program, David R. Cheriton School of Computer Science.*

*Amount: 29,000 CAD / Year*

### McGill Graduation Honours

May 2025

*Distinction & First Class Honours.*

### Science Undergraduate Research Award (SURA)

Summer 2024

*Research funding for full-time research and development activities.*

*Amount: 8,700 CAD*

## Work Experience

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- Teaching Assistant – Intermediate Algorithm Design and Analysis (CPSC 320)** Sept. 2025 – Present  
*The University of British Columbia, Department of Computer Science*  
Vancouver, BC, Canada
- Hold weekly office hours to answer questions from students.
  - Hold weekly tutorials to review class material for 20-30 students.
  - Coordinate a team of undergraduate TAs for test grading.
- Photographer** August 2024 – August 2025  
*Agence Voltaic inc.* Montréal, QC, Canada
- Event photography across all Montreal Universities. View my work at this url: alexstaubinphoto.com
- Course Grader – Honours Linear Algebra (MATH 251)** January 2024 – April 2024  
*McGill University, Faculty of Science, Dept. of Mathematics and Statistics* Montréal, QC, Canada
- Graded the assignments of honours students in MATH 251.
- Intern** June 2023 – December 2023  
*Institut de la Statistique du Québec (ISQ)* Montréal, QC, Canada
- Developed a VBA application which streamlined the process of estimating survey completion times.
  - Sorted and interpreted open-ended responses using word analysis programs.
  - Contributed to provincial research studies such as the *Québec Health Survey of High School Students* (QHSHSS) and the *Québec Survey on Tobacco and Vaping Products*.

## Relevant Projects

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- Dynamic Duos: Robust Duo Adaptation Under Shift** May 2026  
*Poster presented at AI/CRV* SFU
- We investigate better ways to combine a large and small model when the test data drifts away from training distribution.
  - The goal is to take an Asymmetric Duo (a two-model ensemble pairing a large and a small network), and combine the predictions adaptively. To do so, we pull features from both backbones and train a small network to predict per-sample temperatures to weight the ensemble based on how much to trust each model on a given input.
- Adapting PAD to Trifinger Robotic Manipulation in Causal World** Fall 2025  
*CPSC 532X – Adaptation and Adaptive Computation* UBC
- A self-supervised Test-Time Adaptation framework that leverages an auxiliary Inverse Dynamics objective to update the agent's encoder during deployment.
  - Evaluated the method on the CausalWorld robotic benchmark under two distinct shift types: physical dynamics and spatial distribution. Project page.
- TransCoPhy: Modernizing Causal Physical Reasoning with Transformers** Fall 2025  
*CPSC 532Y – Causal Machine Learning* UBC
- Explored the problem of counterfactual physical reasoning from visual input.
  - Modernized the existing CoPhyNet architecture (Baradel et al., ICLR 2020) by replacing its RNN core with a Transformer, creating TransCoPhy. Project page.
- Network Interventions to Limit Misinformation Spread on Reddit** Fall 2025  
*CPSC 534L – Computing and Learning from Graphs* UBC
- Investigated network-based interventions to limit misinformation propagation on Reddit using the FACTOID dataset of users and their interactions. Project page.
- Improving Drotrack** Fall 2024  
*COMP 558 – Fundamentals of Computer Vision* McGill University
- Enhanced the performance of a 2020 drone-based tracking paper by implementing techniques like Kalman filters, optical flow, and ORB features. Code and report available here.

**Machine Learning***COMP 551 – Applied Machine Learning*

Fall 2024

*McGill University*

- Emotion classification using LLMs, Regularization and model evaluation, Classification of image data with MLPs and CNNs. Code and reports available [here](#).

**Image Deblurring***MATH 563 – Honours Convex Optimization*

Winter 2024

*McGill University*

- Deblurred images by implementing convex optimization algorithms such as Spingarn's Method, Primal-dual Douglas Rachford Splitting, ADMM, and Chambolle-Pock in Matlab. Code and report available [here](#).

*Skills*

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**Programming languages:** Python, JAVA, C/C++, MATLAB, VBA, R.**Bilingual:** French, English.**Other :**  $\LaTeX$ , Unix/Linux, Git, ROS, SLURM, SQL, Pytorch, Numpy, Pandas, Excel.